Resolution Geometry

The Quantum Bridge: Task-Level Closure, Backreaction, and Explicit Forward and Inverse Walks

Print edition (supersedes the first edition)

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Abstract

We finalize the passage from quantum observations to resolved geometry as a conditional, task-specific construction. It is not an empirical derivation of quantum gravity. The exact certificate is the constancy of the *final task* on the complete observational fibre, which includes source, coupling, renormalization, boundary and calibration alternatives. Identifying every intermediate link is sufficient but not necessary, so the first edition's "weakest link" rule is demoted to a conservative sufficient rule.

The observation model is repaired. Raw second moments replace centred variances, a quadrature proxy is separated from physical stress, and an explicit conserved cavity-mode stress is derived. Conservation and boundary conditions are imposed, and errors are carried through to resolved geometric tasks. Four walks are executed:

1. an analytic Gaussian walk, forward and backward, with no false structural wall;
2. backreaction, closed rigorously in a two-level feedback analogue and numerically in a trapped Schrödinger–Poisson model;
3. a fourth-moment source task;
4. a finite-shot, dephasing-aware entanglement witness that reaches a negative certified bound on synthetic data.

A 24-check battery passes and rejects eight deliberate mutations. A replay verifier recomputes the reported estimates from saved records, and an independent ten-check audit, sharing no code with the battery, reproduces every key number. These results are mathematical identities and synthetic model tests. They are not experimental data, and they are not a uniqueness theorem for semiclassical gravity.

Reading rule. "Closed" below always means closed for the stated task, model class, domain and error budget. No claim that RG derives the Born rule, Hilbert space, Einstein coupling, or a quantum gravitational mediator is made.

Status vocabulary and placement. Statuses are those of the print ladder [1]: LOCKED, CORRESPONDENCE, ESCALATE, VACUOUS, INCONSISTENT, STRUCTURAL WALL[τ] and OUTSIDE DOMAIN. The bridge relies on the following rungs:

- universal factorization (L2), the confidence-set bridge (L16) and finite-data rank walls (L47–L48);
- contextual quantum protocols and the trace rule (L42–L43), and the non-circularity certificate (L57);

- window-indexed walls and threshold filtrations (L114).

Companion papers supply the classical endpoints: *The Stress Tensor* (geometric and source routes) and *Crossing into Physics* (consilience). The bridge itself is a research programme, not a rung.

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1 The actual bridge and its certificate

Let an admissible model be

$$z = (\rho, \mathcal{I}, \theta_{\mathrm{mat}}, \theta_{\mathrm{ren}}, c, b, \kappa, g),$$

where $\mathcal{I}$ is the measurement-instrument family, θ_{mat} specifies the matter theory, θ_{ren} its renormalization prescription, c the coupling, b boundary/initial data, κ calibration and nuisance parameters, and g a solution of the declared equations. These entries are constrained jointly; their admissible set need not be a Cartesian product. Let $O(z)$ be the predicted observation law and $H(z)$ the final, gauge-invariant task.

Definition 1.1 (Task image)**.** For an exact observation law o, define

$$\mathcal{F}_o = \{z : O(z) = o\}, \qquad \mathcal{Y}_H(o) = \{H(z) : z \in \mathcal{F}_o\}.$$

For finite data D, replace $\mathcal{F}_o$ by a declared confidence/admissibility set $\mathcal{F}_D$. It must include relevant statistical, calibration and model uncertainty.

Theorem 1.2 (Exact and resolution-level closure)**.** *For a nonempty fibre, H descends to observations if and only if*

$$O(z) = O(z') \implies H(z) = H(z').$$

Equivalently, $\mathcal{Y}_H(o)$ is a singleton for every admissible o. For a task metric d_H, define resolution-level forcing by

$$\operatorname{diam}_{d_H} \mathcal{Y}_H(D) \le \tau_H.$$

This is a diameter convention. A radius-τ_H convention would require a different factor and must not be substituted silently.

Proof. If $H = \bar{H} \circ O$, equality of observations implies equality of tasks. Conversely, define $\bar{H}(o)$ as the common task value on $\mathcal{F}_o$; fibre constancy makes the definition independent of representative. The finite-resolution statement is the declared diameter criterion. □

Correction to the original composition claim. If T descends through O and c is fixed, then $c \circ T$ descends. But the converse fails: $O(-1) = O(1)$, $T(\pm 1) = \pm 1$, and $H = T^2 = 1$. Likewise, two geometries may differ while a chosen observable agrees. Thus "the weakest link" is a conservative sufficient rule, not an exact characterization of composite identifiability. The final task image is decisive (F02).

At finite resolution, pairwise closeness is generally not transitive. Do not call $d_H \le \tau_H$ an equivalence relation unless an actual partition or other transitive construction has been supplied. An empty fibre is model incompatibility, not successful identification.

Proposition 1.3 (A usable stability certificate)**.** *Suppose a fixed chain $f_k \circ \cdots \circ f_1$ has uniform Lipschitz constants L_i on the admissible domains. Let the input error be δ_0 and the uniform implementation/model discrepancy of link i be ϵ_i. Then*

$$\delta_k \le \left(\prod_{i=1}^{k} L_i\right)\delta_0 + \sum_{i=1}^{k}\left(\prod_{j=i+1}^{k} L_j\right)\epsilon_i.$$

Proof. Apply $\delta_i \le L_i\delta_{i-1} + \epsilon_i$ recursively. For a coupling class the constants and discrepancies must be uniform over that class; otherwise compute the task image directly. □

For curvature tasks, a small metric amplitude error is insufficient. The sequence $k^{-1}\sin(kx)$ has amplitude k^{-1} and second-derivative amplitude k. A suitable C^2 or Sobolev control, nondegenerate metrics, and bounded inverse metrics are needed (F18). Regularity alone is not a stability bound.

2 Quantum observations: declare the structure rather than derive it

2.1 Contexts and sequential experiments

For a specified quantum model and POVM effects $E_{a|s}$,

$$p(a|s) = \mathrm{Tr}(\rho E_{a|s}).$$

The family $\{p(\cdot|s)\}_s$ is the observation object. A product of marginal distributions is an artificial mathematical coupling, not a joint sharp measurement of incompatible observables. Noncommuting position and momentum do not justify a blanket statement that no joint probability distribution with those marginals exists. A Fine-type result [3] requires its particular compatibility and marginal assumptions.

Sequential tasks need instruments, not just effects:

$$p(a, b|s, t) = \mathrm{Tr}\left[\mathcal{I}_{b|t} \circ \mathcal{I}_{a|s}(\rho)\right].$$

Each outcome map is completely positive and trace-nonincreasing; their sum is trace-preserving. Two instruments can share effects and differ in later statistics. Complete positivity is required when arbitrary ancillas are admitted; positivity on isolated states is insufficient. These are imported quantum operational assumptions [4, 5], not consequences of the RG quotient.

2.2 What a finite observation panel identifies

Theorem 2.1 (Effect-span criterion)**.** *On all density matrices in a finite-dimensional Hilbert space, a Hermitian expectation $\mathrm{Tr}(\rho A)$ is identified by a panel of effects if and only if A lies in the real linear span of those effects and the identity.*

Proof. Sufficiency follows by linearity. Otherwise take a nonzero Hermitian component B of A orthogonal to the span. Orthogonality to the identity makes B traceless. For sufficiently small $\epsilon > 0$, $\rho_\pm = I/d \pm \epsilon B$ are positive density matrices. Their panel probabilities agree, while their A expectations differ by $2\epsilon\,\mathrm{Tr}(BA) \neq 0$. □

The unrestricted-state hypothesis matters: on a restricted state family, nonlinear relations can identify additional tasks. For unbounded field operators, domains and moment bounds must be added; the finite-dimensional theorem is not silently promoted to a QFT theorem (F01).

2.3 The corrected single-mode mean-source panel

With $[a, a^\dagger] = 1$ and $X_\theta = (ae^{-i\theta} + a^\dagger e^{i\theta})/\sqrt{2}$, define the *quadratic target*

$$q_\theta = \langle X_\theta^2 \rangle - \tfrac{1}{2} = n + \Re(ze^{-2i\theta}), \quad n = \langle a^\dagger a \rangle,\ z = \langle a^2 \rangle.$$

It is reconstructed by

$$\begin{aligned} n &= (q_0 + q_{\pi/2})/2, \\ \Re z &= (q_0 - q_{\pi/2})/2, \\ \Im z &= q_{\pi/4} - n. \end{aligned}$$

Three scalar raw second moments suffice and are generically necessary for the unrestricted three-parameter quadratic family. They are *not* three centered variances unless means vanish or are separately measured. Vacuum and a displaced coherent state have the same centered variance 1/2 but different raw moments (F03).

For panel covariance Σ_q, the parameter covariance is $B\Sigma_q B^T$, with B the displayed linear inverse. The matrix is recorded in F03. For unbounded homodyne outcomes, finite-shot confidence requires a tail/fourth-moment assumption, truncation rule or another valid concentration argument. Bounded-Pauli concentration cannot be reused without modification.

Physical-source gate. q_θ is not automatically the complete T_{00} of a field. The oscillator Hamiltonian is $\hbar\omega(n + 1/2)$ and is phase-independent. The negative squeezed value of q_θ alone is not a proof of a negative physical stress component. The field, mode functions, derivatives, subtraction and stress operator must be specified. F04 explicitly separates the proxy from total oscillator energy.

3 From quantum data to a conserved physical source

3.1 Fix the source target

The relevant source is not an arbitrary local energy-density formula. For a specified matter action, the stress is obtained by metric variation with a stated convention; in QFT it is renormalized in a specified scheme. A representative schematic free-field construction is

$$\langle T_{\mu\nu}(x) \rangle_{\mathrm{ren}} = \lim_{x' \to x} D_{\mu\nu}(x, x') \left[W(x, x') - H_{\mathrm{sub}}(x, x') \right] + C_{\mu\nu}[g].$$

Here W is the two-point function, H_{sub} a suitable local subtraction, and $C_{\mu\nu}$ the fixed finite local counterterms. Hadamard-type state regularity, domains, conservation and the prescription are hypotheses. Renormalization ambiguities may include conserved local curvature tensors [6, 7]; they must be fixed or retained as model alternatives.

For a quadratic free-field stress, a sufficient two-point function can identify the mean stress. A single mode's three moments do not identify an arbitrary multimode stress field. Interactions can require higher correlations. Conversely, a position density does not identify general stress: multiplying a Gaussian amplitude by e^{ikx} preserves its position law but changes momentum and kinetic energy.

3.2 An explicit physical mode: the missing source step

Here the quadrature target is connected to an actual stress contribution rather than simply renamed. In units $\hbar = c = 1$, take a minimally coupled massless real scalar in a flat cavity of

length L, periodic transverse area $A_\perp$, and Dirichlet ends. Retain one standing mode,

$$f(x) = \sqrt{\frac{2}{A_\perp L}}\sin(\omega x), \quad \omega = \frac{\pi}{L}, \qquad \phi(t,\mathbf{x}) = \frac{f(x)}{\sqrt{2\omega}}(ae^{-i\omega t} + a^\dagger e^{i\omega t}).$$

Let $z_t = ze^{-2i\omega t}$. The normal-ordered contribution relative to the specified cavity vacuum is

$$\begin{aligned}
\Delta T^{00} = \Delta T^{xx} &= \frac{\omega}{A_\perp L}\left[n + \cos(2\omega x)\Re z_t\right], \\
\Delta T^{0x} = \Delta T^{x0} &= -\frac{\omega}{A_\perp L}\sin(2\omega x)\Im z_t, \\
\Delta T^{yy} = \Delta T^{zz} &= -\frac{\omega}{A_\perp L}\left[n\cos(2\omega x) + \Re z_t\right].
\end{aligned}$$

All other components vanish. Direct differentiation gives

$$\partial_t \Delta T^{00} + \partial_x \Delta T^{x0} = 0, \qquad \partial_t \Delta T^{0x} + \partial_x \Delta T^{xx} = 0$$

in the cavity interior, and $\int \Delta T^{00} d^3x = \omega n$. Thus the three raw moments genuinely determine this physical mode contribution, including its flux and pressures. F24 verifies both conservation identities and the integrated energy symbolically.

For a squeezed state with $r = 0.6$, $n = \sinh^2 r$, $z = -\sinh r \cosh r$, the contribution at $t = 0$, $x = L/12$ is proportional to

$$\sinh^2 r - \frac{\sqrt{3}}{2}\sinh r \cosh r < 0.$$

This is a negative *normal-ordered contribution*, not by itself a proof that the full absolute renormalized energy density violates a particular energy condition. The reference-vacuum stress, other modes and wall/support stress must be added. Interior conservation does not remove the force on the cavity ends. These qualifications are part of the bridge, not corrections to be hidden after coupling.

For this specified mode, stress is a linear function of $(n, \Re z, \Im z)$. Its covariance follows from the panel inverse and those coefficient functions. If the mode functions, background or apparatus are unknown, their uncertainty must also enter the admissible source set.

3.3 Local-density and Casimir examples, properly scoped

For a real one-dimensional wavefunction, the two familiar kinetic densities obey

$$t_1 = \frac{\hbar^2}{2m}|\psi'|^2, \quad t_2 = -\frac{\hbar^2}{2m}\Re(\psi^*\psi''), \quad t_1 - t_2 = \frac{\hbar^2}{4m}(|\psi|^2)''.$$

Their integrals agree under suitable boundary conditions. This is a local-density ambiguity, not permission to choose any gravitational source. The action, improvement terms and boundary terms decide which stress is coupled.

For ideal parallel plates, the declared bulk symmetries and tracelessness, together with the normal pressure, yield

$$T^{\mu\nu} = \frac{\pi^2\hbar c}{720a^4}\,\mathrm{diag}(-1,1,1,-3).$$

Without tracelessness, the parallel block coefficient remains free. This is conditional ideal-model identification [8, 9]; force measurements report agreement at the 15%, 5% and 1% levels in different geometries [10, 11, 12]. The plate/support apparatus, material corrections, boundary stresses and gravitational response of the complete apparatus are separate tasks. The isolated negative interaction energy divided by c^2 is not a complete weighing prediction for an apparatus.

3.4 Conservation and admissibility

The gravitational equations require a compatible total source:

$$\nabla^{\mu} T_{\mu\nu}^{\text{total}} = 0.$$

The total includes traps, supports and other relevant matter. Bianchi identities make this a consistency condition, not an optional numerical diagnostic. A prescribed static pressure gradient without balancing forces fails the gate (F17). Quantum energy inequalities [13] and energy conditions are distinct notions. Any energy-condition hypothesis must be checked for the actual renormalized source and regime; it is neither automatic nor merely a relabeling of the coupling.

3.5 A full linearized propagation theorem

A general source-to-geometry link can now be stated without a weakest-link shortcut. In units $c = 1$, fix Minkowski background, harmonic gauge, a retarded/no-incoming prescription and any homogeneous initial solution. Suppose the *complete* conserved source, including apparatus where required, has the known form

$$T_{\mu\nu} = T_{\mu\nu}^{\text{ref}} + nU_{\mu\nu} + (\Re z)V_{\mu\nu} + (\Im z)W_{\mu\nu},$$

with coefficient tensors of suitable support and regularity. These are hypotheses on the completed physical model; the cavity-interior calculation alone does not establish them for the walls.

Theorem 3.1 (Quadratic-source to linearized-geometry descent)**.** *Under these assumptions, the three raw second-moment panel identifies the retarded linearized metric modulo the fixed gauge convention, and every specified gauge-invariant linear geometric task of that metric. Its uncertainty is the pushforward of the full panel/calibration/model uncertainty, not a new independently chosen error bar.*

Proof. In harmonic gauge, $\Box \bar{h}_{\mu\nu} = -16\pi G T_{\mu\nu}$. The fixed retarded solution is

$$\bar{h}_{\mu\nu}(t, \mathbf{x}) = 4G \int \frac{T_{\mu\nu}(t - |\mathbf{x} - \mathbf{y}|, \mathbf{y})}{|\mathbf{x} - \mathbf{y}|}\, d^3y + \bar{h}_{\mu\nu}^{\text{hom}}.$$

Conservation supplies the compatible gauge constraint. Linearity gives a known affine function of $(n, \Re z, \Im z)$. Trace reversal recovers h; a fixed linear geometric task preserves the affine dependence. The panel identifies these three coefficients, so the task is constant on its observational fibres. A bounded linear response on the chosen norms propagates uncertainty by the corresponding linear map; otherwise its stability must be proved separately. □

This theorem is the formal bridge, not a derivation of Einstein coupling. It explicitly exposes the physical completion, boundary and stability assumptions. The numerical walk below executes a static Newtonian specialization; no claim is made that the full time-dependent cavity/apparatus integral was numerically solved here.

4 Walk I: quantum position law to weak-field geometry and back

4.1 Declare the model

Assume a nonrelativistic, stationary, spherically symmetric mass distribution with known centre, mass M and Gaussian position width σ:

$$\rho_m(r) = \frac{M}{(2\pi\sigma^2)^{3/2}} e^{-r^2/(2\sigma^2)}.$$

The rest-mass density is the leading Newtonian source. This is not a claim to have reconstructed its full relativistic stress or quantum state. If a trap prepares the state, its stresses must be included at the order where they matter. Assume weak gravity and nonrelativistic velocities; retain only the leading Newtonian sector. Fix $\Phi(\infty) = 0$, regularity at the origin, the coordinate/gauge convention, and the calibration of the potential task.

4.2 Forward map and all-component check

The unique decaying regular solution in this model is

$$\nabla^2\Phi = 4\pi G\rho_m, \qquad \Phi(r) = -\frac{GM}{r}\,\mathrm{erf}\left(\frac{r}{\sqrt{2}\sigma}\right).$$

The value at the origin is understood by continuity. With $x^0 = ct$,

$$ds^2 = -(1 + 2\Phi/c^2)(dx^0)^2 + (1 - 2\Phi/c^2)d\mathbf{x}^2.$$

For a general static Φ, the linearized mixed Einstein tensor is

$$G^{(1)0}{}_0 = -\frac{2}{c^2}\nabla^2\Phi, \qquad G^{(1)0}{}_i = G^{(1)i}{}_j = 0.$$

Using $T^0{}_0 = -\rho_m c^2$ returns the source through $G^0{}_0 = 8\pi G T^0{}_0/c^4$. F05 checks all components symbolically, independently of the original spherical calculation; F06 checks the Gaussian Poisson identity. The identities are exact *within the linearized model*, not exact nonlinear Einstein solutions.

Without boundary conditions one may add a harmonic function. For example, adding ax changes acceleration while leaving the Laplacian unchanged. Source descent therefore does not by itself identify geometry (F07).

4.3 Finite-data walk

For N independent centred three-dimensional Gaussian samples,

$$\widehat{\sigma}^2 = \frac{1}{3N}\sum_{i=1}^{N}|\mathbf{x}_i|^2, \qquad \frac{\sum_i |\mathbf{x}_i|^2}{\sigma^2} \sim \chi^2_{3N}.$$

Known centre and independent ideal position measurements are explicit assumptions. For the synthetic run with $N = 20{,}000$, true $\sigma = 1$ and seed 20261021:

$$\begin{aligned}
\widehat{\sigma} &= 1.0007307413,\\
I_{\sigma,95\%} &= [0.9951007665,\ 1.0064252296],\\
\widehat{\Phi}(1)/(GM) &= -0.6823359853,\\
I_{\Phi(1)/(GM),95\%} &= [-0.6850662421,\ -0.6795900463].
\end{aligned}$$

The interval is propagated monotonically through the exact model map (F10). Its coverage is conditional on the sampling model, not a guarantee for one realized interval. The delta-method relative standard error of $\widehat{\sigma}$ is $1/\sqrt{6N} = 0.00288675$. Mass and calibration uncertainty, if present, must also be propagated; they are held fixed here.

4.4 Inverse map: no false structural wall

A Gaussian has infinite support. There is no finite radius outside all its matter. At every finite $r > 0$,

$$\frac{\partial \Phi}{\partial \sigma} = \frac{GM}{\sigma^2}\sqrt{\frac{2}{\pi}}\, e^{-r^2/(2\sigma^2)} > 0.$$

Consequently its width is not exactly invisible. At $r = 5\sigma$,

$$\frac{\partial \log|\Phi|}{\partial \log \sigma} = -1.4867203671 \times 10^{-5}.$$

This is ill-conditioning. Whether it becomes a resolution wall depends on the width range, task metric and error budget, not the derivative alone.

Two exact potential values at radii 1 and 2 recover the synthetic $M = 1.3$, $\sigma = 1.2$ to numerical precision (F09). The relative Jacobian for parameters $(\log M, \log \sigma)$ has condition number about 5.095 at radii $(1, 2)\sigma$, versus 1.352×10^5 at $(5, 6)\sigma$. These are local sensitivity comparisons, not global inverse guarantees for arbitrary profiles.

For compact, spherically symmetric sources with the same total mass, the common exterior potential really is $-GM/r$. Their radial profiles then form an exact observational fibre. This is the correct shell-theorem structural ambiguity. Neither case reconstructs a quantum phase, a general density matrix or an arbitrary interior stress.

5 Walk II: close the backreaction loop rather than assume it

5.1 Conditional fixed-point theorem

On a resolved background the quantum source may depend on geometry:

$$g \;\mapsto\; \rho_g \;\mapsto\; T_{\mathrm{ren}}[g, \rho_g] \;\mapsto\; \mathcal{S}_{c,b}(T) =: \mathcal{F}(g).$$

In general this is an implicit coupled problem, not a one-way map. Gauge, state preparation, initial/boundary conditions, renormalization and matter conservation must all be fixed or retained in the admissible set.

Theorem 5.1 (A sufficient feedback closure; Banach [14])**.** *Let $\mathcal{F}$ map a complete closed admissible metric space into itself and satisfy $d(\mathcal{F}g, \mathcal{F}h) \le q\, d(g, h)$ with $q < 1$. Then it has a unique fixed point, iteration converges, and*

$$d(g, g_*) \le \frac{d(g, \mathcal{F}g)}{1 - q}.$$

Proof. Successive iterate distances decrease geometrically, so the iterates are Cauchy and converge in the admissible set. Continuity gives a fixed point; contraction gives uniqueness. Finally $d(g, g_*) \le d(g, \mathcal{F}g) + q\, d(g, g_*)$. □

This is a sufficient theorem, not a claim that the full semiclassical Einstein–QFT map satisfies its hypotheses. Multiple solutions must remain a correspondence unless a selection principle is declared and justified.

5.2 A fully walked quantum feedback analogue

Consider the dimensionless two-level Hamiltonian

$$H(g) = \Delta\sigma_x + g\sigma_z, \quad \rho_g = \tfrac{1}{2}\left[I - \frac{\Delta\sigma_x + g\sigma_z}{\sqrt{\Delta^2 + g^2}}\right], \quad \Delta > 0.$$

Its ground-state source is $t(g) = \mathrm{Tr}(\rho_g\sigma_z) = -g/\sqrt{\Delta^2 + g^2}$. Declare the feedback law $g = b + \lambda t(g)$ with $\lambda \geq 0$. The map

$$\mathcal{F}(g) = b - \frac{\lambda g}{\sqrt{\Delta^2 + g^2}}$$

maps $[b - \lambda, b + \lambda]$ into itself and has derivative magnitude at most λ/Δ. Thus $\lambda < \Delta$ closes this loop rigorously. For $\Delta = 1$, $b = 0.3$, $\lambda = 0.2$, F21 gives

$$g_* = 0.2512624377, \qquad t(g_*) = -0.2436878115,$$

with fixed-point residual 5.55×10^{-17}. This is a quantum-source feedback analogue, not a spacetime gravitational model. It demonstrates the missing logical step without pretending to prove it for full gravity.

5.3 A gravitational mean-field numerical walk

Declare instead the dimensionless trapped Schrödinger–Poisson model

$$\left[-\tfrac{1}{2}\frac{d^2}{dr^2} + \tfrac{1}{2}r^2 + \lambda\phi_u(r)\right]u = \epsilon u, \qquad \int_0^\infty u^2 dr = 1,$$
$$\phi_u(r) = -\left[\frac{1}{r}\int_0^r u(s)^2 ds + \int_r^\infty \frac{u(s)^2}{s} ds\right].$$

Here $u = \sqrt{4\pi} r\psi$, and $\lambda = 0.2$. This is an explicitly chosen nonrelativistic mean-field coupling, of Schrödinger–Newton type [15, 16]. It is not a derivation of single-particle self-gravity from RG, and its physical status is itself contested. The trap is externally prescribed; this model is not a full relativistic apparatus solution.

The executable uses $r \in (0, 8)$, Dirichlet endpoints, a tridiagonal ground-state solve and damped iteration. At 800 interior points it obtains

$$\epsilon = 1.3386825453, \quad \sqrt{\langle r^2\rangle} = 1.2083361267, \quad \|Hu - \epsilon u\|_h = 3.78 \times 10^{-12}.$$

Here ϵ is the nonlinear eigenvalue, not the total Hartree energy. The 400-point eigenvalue is 1.3386279146, a difference 5.46×10^{-5}. The probability beyond $r = 6$ on the fine grid is about 1.21×10^{-15} (F11). The discretized residual, tail diagnostic and two-grid comparison are numerical evidence, not a rigorous continuum error bound or a uniqueness proof. In particular, inner-cell quadrature and finite-box errors remain numerical approximations.

6 Walk III: extend source descent beyond means

6.1 A concrete fourth-moment task

The Fock state $|1\rangle$ and a thermal state with mean occupation one have identical $n = 1$, $z = 0$, but trace distance 3/4 and number variances 0 and 2. Mean-source identification does not identify the state or its fluctuations.

For a canonical single mode,

$$\frac{1}{\pi}\int_0^{\pi}\langle X_\theta^4\rangle d\theta = \tfrac{3}{2}\langle n^2\rangle + \tfrac{3}{2}\langle n\rangle + \tfrac{3}{4}.$$

The fourth moment contains only phase harmonics $0, \pm 2, \pm 4$. Five equally spaced phases $\theta_k = k\pi/5$ reproduce the phase average exactly. Hence

$$\langle n^2\rangle = \frac{2}{15}\sum_{k=0}^{4}\langle X_{k\pi/5}^4\rangle - \langle n\rangle - \tfrac{1}{2}.$$

This is a sufficient panel, not a minimality claim for the number-variance task. F22 reconstructs $\langle n^2\rangle = 9.4523850852$ with absolute error below 10^{-11} on a synthetic finite-support state. Finite Fock cutoffs do not obey the canonical commutator at the top state; the test keeps state support away from that boundary.

6.2 A fluctuation-level geometry map needs more structure

For physical stress, define smeared observables and the symmetrized connected noise kernel [17]

$$N_{\mu\nu\alpha\beta}(x,y) = \tfrac{1}{2}\langle\{\Delta T_{\mu\nu}(x), \Delta T_{\alpha\beta}(y)\}\rangle.$$

Products at coincident points require distributional/renormalization control. A one-time fourth-moment panel is not automatically a complete spacetime noise kernel. In a declared linear stochastic-response model, write

$$h = K\xi + h_0, \qquad C_h = KNK^* + C_0 + KC_{\xi 0} + C_{0\xi}K^*.$$

Cross terms vanish only under a stated uncorrelatedness assumption. Detector covariance is then added for the observed task. F12 checks a finite-dimensional uncorrelated example with C_0 retained.

The response K, its retarded boundary conditions, dissipation and any intrinsic noise are coupling inputs. Symmetrized covariance alone does not determine quantum commutators, causal response or a quantum channel. This walk closes a declared fluctuation model, not the quantum nature of gravity.

6.3 Mean-field versus outcome-conditioned alternatives

For mean-field [18, 19] and outcome-conditioned couplings the following holds. If a field equation is linear, both couplings use the same boundary data, and a task is linear, then the task evaluated on the mean field equals its outcome average. The macroscopic regime was probed by Page and Geilker [20]. This does not extend to nonlinear tasks or nonlinear Einstein evolution. For outcomes ± 1, the mean is zero while the mean square is one (F14).

For the supplied two-branch acceleration toy model, $M = 1$ kg, $R = 0.1$ m,

$$s_a(d) = \frac{GMd}{2(R^2 + d^2/4)^{3/2}}.$$

At the *declared hypothetical* resolution 10^{-15} m s^{-2}, the small-separation threshold is $d \simeq 2.99657 \times 10^{-8}$ m. In particular $d = 10^{-6}$ m gives $s_a = 3.33715 \times 10^{-14}$ m s^{-2} and is above that threshold; 10^{-9} m gives 3.33715×10^{-17} and is below (F13). The original prose incorrectly placed the micrometre case below threshold. Standard deviation versus detector resolution is only this toy comparison, not a universal statistical distinguishability criterion or an instrument-performance claim.

7 Walk IV: a finite-shot entanglement discriminator

7.1 State, alternatives and assumptions

Use the ideal two-path Newtonian phase model of the gravitational-entanglement proposals [21, 22], initially $|+\rangle|+\rangle$, with

$$\phi_{ij} = \frac{GM^2 t}{\hbar r_{ij}}, \quad |\Psi\rangle = \tfrac{1}{2}\sum_{i,j} e^{i\phi_{ij}}|ij\rangle, \quad \Delta\phi = \phi_{00} + \phi_{11} - \phi_{01} - \phi_{10}.$$

For $M = 10^{-14}$ kg, path separation 250 μm, closest separation 200 μm and $t = 2.5$ s, the script obtains

$$\Delta\phi = -0.3139344926, \qquad C_{\text{ideal}} = |\sin(\Delta\phi/2)| = 0.1563234613.$$

These are model inputs and predictions, not experimental observations or verified current proposal specifications.

Local operations and classical communication preserve separability: each product Kraus branch maps a product state to a product state, and classical mixtures remain convex mixtures of products. Thus a certified entangled output from an initially separable state excludes that LOCC class. A single product-phase numerical example is not a proof of this theorem; F16 supplies only implementation controls.

Interpretation gate. Identifying gravity as the cause additionally requires controlled non-gravitational interactions, preparation, readout, postselection, locality/mediation assumptions and backgrounds. A direct entangling interaction model need not specify a mediator Hilbert space. Entanglement alone neither uniquely identifies a coupling nor establishes a complete quantum theory of gravity. A null result also does not establish a classical mediator.

7.2 Dephasing and an executable witness

Apply independent local phase damping with visibility v:

$$(\rho_v)_{ij,kl} = v^{[i\neq k]+[j\neq l]}(\rho_{\text{ideal}})_{ij,kl}.$$

In this specified two-qubit family, the partial-transpose [23, 24] boundary occurs at $v \simeq 0.8558213033$. This threshold is model-specific. Choose the negative-eigenvalue eigenvector $|\eta\rangle$ of $\rho_{\text{ideal}}^{T_B}$ and register

$$W = (|\eta\rangle\langle\eta|)^{T_B} = \sum_{\mu,\nu=0}^{3} w_{\mu\nu}\sigma_\mu \otimes \sigma_\nu.$$

For every separable state $\mathrm{Tr}(W\rho) \geq 0$. The witness is fixed from the declared model before collecting the synthetic observations. At $v = 0.99$, its model expectation is -0.0724051134.

Nine local Pauli-pair settings estimate the 15 nonidentity Pauli expectations, including marginals. Let $L = \sum_{(\mu,\nu)\neq(0,0)} |w_{\mu\nu}| = 1.0034447569$. With n independent shots per setting, Hoeffding's inequality [25] and a union bound over the *15 estimated quantities*, not merely the nine settings, give the sufficient bound

$$n \geq \frac{2L^2}{\epsilon_{\text{stat}}^2} \log\frac{30}{\alpha}.$$

No independence between the 15 estimates is needed for the union bound; independent bounded shots within each setting are assumed. The bound is conservative, not an optimized power analysis.

Set $\alpha = 0.05$, $\epsilon_{\text{stat}} = 0.02$ and an assumed valid systematic bound $\epsilon_{\text{sys}} = 0.005$. This gives $n = 32{,}206$ per setting, 289,854 total shots. F23 actually samples the nine multinomial setting laws with seed 20261022 and obtains

$$\widehat{w} = -0.0691516395, \qquad \widehat{w} + \epsilon_{\text{stat}} + \epsilon_{\text{sys}} = -0.0441516395 < 0.$$

The synthetic walk therefore reaches a negative confidence upper bound under the declared model and systematic allowance. It is not an empirical gravitational-entanglement result. The assumed systematic bound must itself be justified in an experiment; no environmental background budget has been measured here.

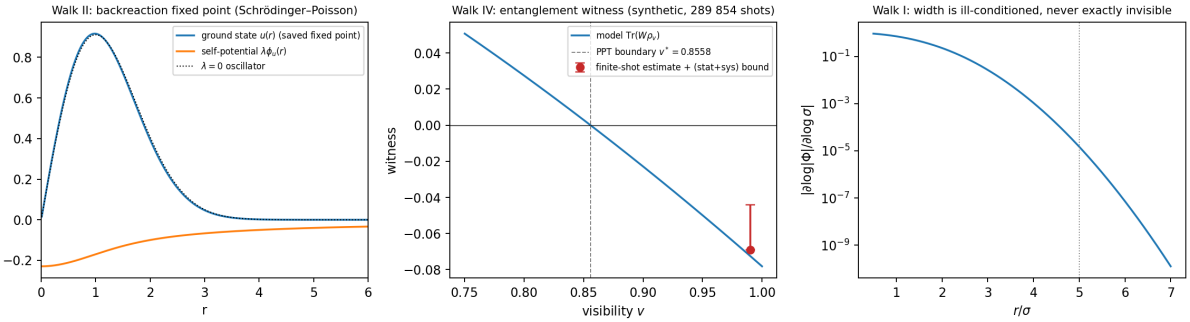


Figure 1: Three walks, drawn from the saved records (no regeneration) and from analytic curves. Left: the Schrödinger–Poisson fixed point ($\lambda = 0.2$) and its self-potential, against the $\lambda = 0$ oscillator. Centre: the registered witness against the dephasing visibility, with the PPT boundary and the finite-shot estimate with its (statistical plus systematic) upper bound at $v = 0.99$. Right: sensitivity of the potential to the Gaussian width. It is ill-conditioned at large radius but never zero.

8 Errata to the first edition

The first edition [2] overstated or misstated the following points. Each is corrected above.

1. *Composition.* "The composite has the status of its weakest link" was presented as a characterization. It is a sufficient rule. The exact certificate is the final task image (Theorem 1.2; F02).

2. *Contexts.* "Position and momentum have no joint law (Fine)" was a blanket overstatement. A product of marginals is a joint distribution with those marginals. Fine-type theorems need their particular compatibility assumptions.

3. *Panel.* "Three homodyne variances" should read three *raw* second moments. Centred variances lose the mean (F03).

4. *Proxy and stress.* The single-mode quadrature target was called a renormalized energy density. It is a proxy. The physical mode stress needs the field, the mode functions and the subtraction (F04, F24).

5. *Energy conditions.* A negative normal-ordered contribution was read as a violation of the energy condition. The full renormalized source, including apparatus, must be assessed.

6. *Inverse wall.* The Gaussian width was called a STRUCTURAL WALL. A Gaussian has infinite support, so the width is ill-conditioned but never exactly invisible (F08). The exact shell-theorem fibre exists for compact sources.

7. *Threshold.* The micrometre superposition was classified FORCED. Its spread is $3.34 \times 10^{-14}\,\mathrm{m\,s^{-2}}$, above the declared 10^{-15}, so it is CONDITIONAL (F13). The error came from a rounding helper that erased small magnitudes, and it propagated into the prose.

8. *Separability.* A single product-phase example was presented as if it proved the LOCC bound. The theorem is the separability-preservation argument, and the example is only a control (F16).

9. *Casimir weighing.* The interaction energy divided by c^2 is not a complete prediction of how much the apparatus weighs.

9 Final bridge status and replacement claims

Component	Defensible final statement
Quantum operational model	Hilbert/Born/POVM/instrument structure is declared. RG classifies tasks within it; it does not derive that structure.
Mean-source descent	Exact for specified targets in the panel span, or for a declared quadratic field model with sufficient two-point data. Not universal tomography.
Physical stress	Conditional on action, subtraction, finite counterterms, domains, conservation and apparatus model. Local-density formulas are not freely interchangeable gravitational sources.
Mean geometry	Conditional on coupling, gauge, boundary/initial data, existence and task stability. The Gaussian Newtonian/linearized walk is explicit.
Backreaction	A contraction theorem closes the two-level analogue. A trapped Schrödinger–Poisson fixed point is numerically demonstrated. Full Einstein–QFT existence, uniqueness and stability are not established.
Inverse geometry	Gaussian width is nonzero but weakly visible at finite radius. Compact spherical exterior profiles have a genuine exact ambiguity. General quantum-state recovery is not shown.
Fluctuation geometry	Requires higher correlations, smearing, response, intrinsic noise and cross-correlation assumptions. A declared linear covariance bridge is walked.
Entanglement task	A dephasing-aware, finite-shot synthetic witness excludes the separable/LOCC null within its assumptions. It does not empirically select a gravitational mediator or a unique coupling.
Composite certificate	Singleton or sufficiently small final task image on the full admissible fibre. Intermediate nonidentification need not survive task projection.

Final characterization. The bridge is now a task-level inference construction with explicit source descent, physical-source gates, coupling and boundary declarations, feedback criteria, forward/inverse walks and finite-resolution certificates. Its conditional mathematical content is closed where the stated hypotheses and tests apply. Empirical coupling selection and general curved-background semiclassical closure remain open obligations, not gaps concealed by a passing test battery.

A Executable audit and reproducibility

`rg_qm_bridge_final_checks.py` runs without network access using NumPy, SciPy and SymPy. Run:

```
python rg_qm_bridge_final_checks.py \
  --out rg_qm_bridge_final_results.json
```

The output records the script SHA-256, numerical evidence and mutation results. A companion NPZ stores the actual synthetic position samples, Pauli-setting counts, witness coefficients and radial fixed-point profile. The independent replay script recomputes the reported estimates and residuals from those saved records. The 24 checks cover: effect-span counterstates; task projection; raw-moment inversion; proxy/energy distinction; all-component linearized Einstein identities; Gaussian Poisson closure; boundary ambiguity; Gaussian sensitivity; two-radius inversion; finite-sample propagation; mean-field feedback; covariance transport; corrected acceleration threshold; nonlinear mean countertasks; noisy witness budget; separable controls; conservation; derivative instability; residual bounds; state/source distinction; contractive quantum feedback; fourth-moment descent; an executed finite-shot witness; and an explicit conserved scalar-mode stress.

All eight deliberate mutations are rejected, and the corresponding unmutated acceptance gates are checked first: exact Gaussian invisibility; micrometre invisibility; omission of intrinsic noise; ignoring dephasing; necessity of intermediate identification; wrong panel inverse; wrong Poisson potential; and omission of feedback. These controls are not an exhaustive adversarial validation.

The independent audit `rg_qbf_audit.py` (H01–H10) and the figure script `rg_qbf_figure.py` complete the bundle; the figure reads only the saved records. The original supplied QB01–QB10 script was rerun unchanged and passed 10/10. That result is preserved as a historical baseline, *not* as validation of its stronger prose claims. The available earlier closure-check script was also rerun, passing 20/20. The final 24-check battery is a separate revised artifact; the counts are not pooled into a misleading independent-test total.

B Independent audit

A separate script, `rg_qbf_audit.py` (H01–H10), shares no code with the battery. It recomputes the key claims:

- the cavity-mode stress components, both conservation laws and the integrated energy ωn, derived from $T^{\mu\nu} = \partial^\mu\phi\,\partial^\nu\phi - \frac{1}{2}\eta^{\mu\nu}(\partial\phi)^2$ for a normal-ordered single mode;
- the negative squeezed contribution;
- the PPT boundary $v^* = 0.8558213033$ and the witness value -0.0724051134;
- $L = 1.0034447569$ and $n = 32{,}206$ shots per setting;
- the fourth-moment panel (error 3×10^{-14});
- the threshold $d^* = 2.99657 \times 10^{-8}\,\mathrm{m}$, with the micrometre case CONDITIONAL;
- the width sensitivity -1.48672×10^{-5} and the condition numbers 5.095 and 1.352×10^5;
- the composition counterexample;
- the feedback fixed point;

- the $|1\rangle$/thermal distinction.

All ten pass. One audit check (H07) first used a central difference of $\log|\Phi|$ near erf ≈ 1. Cancellation gave a relative error of 9×10^{-5}; the analytic form reproduces the paper's value. The discrepancy was in the audit, not in the paper.

C Claim discipline

This edition deliberately replaces, rather than repeats, the unsupported implications of the first:

- a passing finite battery is not a theorem;
- sampled systems are not exhaustive enumeration;
- a quadrature proxy is not a complete field stress;
- equality of means is not equality of laws;
- low sensitivity is not exact nonidentifiability;
- conditional model closure is not empirical quantum gravity.

All numerical walks are newly executed mathematical or synthetic-model calculations. No external experiment, current sensitivity limit or status of current gravitational-entanglement experiments is claimed. Literature values are cited for context only.

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